

A Causal Evaluation of PCA-Guided Sensor-Rate Allocation under Hard Communication Budgets

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Abstract

Purpose: We examine whether incremental principal component analysis (PCA) covariance contributions provide a useful unsupervised signal for allocating sensor transmission rates under a hard communication budget. The study focuses on causal timing and the relationship between measurement distortion and downstream detection.

Methods: Normalized PCA participation scores determine next-window integer quotas. Sampling decisions use only preceding reconstructed windows, with uniform initialization and causal zero-order hold. We evaluate independent Tennessee Eastman process (TEP) simulation runs, Server Machine Dataset (SMD-1-1), and Pooled Server Metrics (PSM) under retrained and fixed detectors, using five joint experimental seeds. Acquisition masks supply actual sample counts; payload bytes assume four bytes per value. Event metrics preserve run boundaries, and predeclared training-score quantiles assess threshold sensitivity.

Results: At 50% bandwidth with retrained detectors, PCA-guided allocation obtains TEP weighted F1 0.675, compared with 0.707 for Uniform and 0.728 for full data. SMD-1-1 and PSM additionally expose dependence on the downstream detector and threshold. No universal superiority is claimed.

Conclusion: PCA covariance participation is a limited allocation proxy. A causal evaluation must distinguish measurement fidelity, diagnostic utility, and communication accounting. The results support a reproducible assessment of this rule rather than universal superiority or embedded deployment readiness.

Keywords: adaptive sampling; sensor-rate allocation; incremental PCA; communication constraints; causal reconstruction; edge measurement

1 Introduction

Distributed measurement systems often collect more channel samples than a sensor link or edge uplink can transmit. Uniform downsampling is simple, but it does not account for differences among channels. A rate allocator can instead assign a rate r_j to each of d channels while respecting a total budget B . The decision is operationally meaningful only if it is made before the samples governed by that decision are observed.

Prior work includes event-triggered temporal sampling [1, 2], correlation-aware sensor selection [3, 4], and online subspace tracking [5–7]. These areas address related questions, but binary channel selection, uniform temporal sampling, and proportional per-channel rate allocation are not interchangeable. The term *sensor triage*, used in an earlier version of this work, means only proportional sensor-rate allocation here.

This study asks whether the channel contributions to an online low-rank covariance approximation provide a useful unsupervised signal for allocating a hard communication budget. PCA itself is not presented as novel. The specific contribution is a causal mapping from incremental PCA covariance contributions to next-window integer sampling quotas, together with an evaluation protocol that exposes its measurement and detection trade-offs.

The contributions are:

1. a causal system and timing model in which rates for window t depend only on information available through window $t - 1$;
2. a dimensionally consistent score equal to the diagonal of the rank- k PCA covariance approximation;
3. a largest-remainder quota rule that enforces the aggregate budget in each window rather than only in expectation; and
4. an evaluation based on independent runs or official chronological splits, with distortion, communication, point-level, event-level, and delay metrics.

2 Related work

Table 1 positions the allocation rule against related methods.

Send-on-delta triggers transmission when the current observation differs sufficiently from the previous transmitted value [1]. This information pattern is natural for a locally observed sensor but differs from deciding an acquisition schedule before values arrive. Bacciu’s cross-correlation filter selects nonredundant time-series features [3]. Ghosh et al. select an active sensor set using upper-confidence-bound learning and energy information, and reconstruct inactive parameters with Gaussian process regression [4]. These methods address relevance or redundancy more directly than a marginal covariance-diagonal score.

A particularly close predecessor is the PCA-SBL framework of Gupta and De [2]. It combines PCA-based sparse signal recovery, energy-aware sensor selection, and feedback that predicts the next measurement cycle’s sampling requirement. Consequently, neither combining PCA with adaptive sensing nor making a next-cycle decision is novel by itself. The present study instead isolates a simpler proportional covariance-participation rule under an explicit integer sample cap and common causal

Table 1 Relationship to the closest methodological families. “Rate” means proportional per-channel rates rather than binary inclusion.

Approach	Decision	Main signal	Strength	Difference from this study
Send-on-delta [1]	transmit on change	local deviation	event responsiveness	observes current value before deciding
Bacciu [3]	select feature subset	cross-correlation	redundancy filtering	does not assign temporal quotas
Ghosh et al. [4]	active sensor set	UCB, correlation, energy	predicts inactive sensors	energy-aware selection with GPR
Gupta and De [2]	active set and cycle rate	PCA-SBL recovery	feedback on sensing error	selection/recovery optimization
CCIPCA [5]	update subspace	covariance structure	incremental tracking	supplies no allocation rule
Present rule	next-window quotas	covariance diagonal	explicit sample cap	descriptive unsupervised proxy

reconstruction. It does not establish superiority over these complete selection and recovery systems.

Recent sensor-sampling work extends this landscape. CLARA combines node correlation, clustering, sampling-rate adaptation, and fault tolerance in periodic wireless sensor networks [8]. Its network-oriented scope motivates correlation-aware allocation, but our replay experiments do not model network clustering or node failures. CoSS jointly selects sensors and sampling rates using weights learned within a human-activity recognition network [9]. This is a task-trained selection mechanism; the present allocator uses no class labels to choose quotas. Huang et al. formulate multi-sensor sampling as a Markov decision process and train a deep Q-network using a reward that combines information, energy cost, and redundancy [10]. Their formulation highlights the importance of specifying both the information available at decision time and the resource objective. These recent methods are relevant alternatives, not methods against which the present results establish an empirical ranking.

Online subspace methods such as Oja’s rule, CCIPCA, and GROUSE can replace the incremental SVD update [5, 6, 11]. They change subspace estimation and computational cost, but do not by themselves specify a communication policy. The method studied here supplies no new process-monitoring statistic or control limit; downstream detectors are measurement consumers used to evaluate the allocation rule.

3 Method

3.1 System, budget, and information pattern

Let $X_t \in \mathbb{R}^{w \times d}$ denote the full but initially unobserved measurements in window t . Before X_t arrives, a controller announces rates $r_{t,j} \in [r_{\min}, 1]$. The sensing or transmission mechanism retains a subset Ω_t and the receiver observes only $P_{\Omega_t}(X_t)$. The aggregate hard constraint is

$$|\Omega_t| \leq \lfloor Bwd \rfloor. \quad (1)$$

The controller reconstructs $\hat{X}_t$ and then updates the PCA state. Thus r_t is measurable with respect to reconstructed observations through $t-1$ and cannot depend on unseen values in X_t .

The first window is a uniform bootstrap, $r_{1,j} = B$. This cost is included in all reported communication measurements. The strict acquisition-constrained model updates PCA with $\hat{X}_t$. A separate uplink-constrained variant, in which a local gateway legitimately observes full-rate X_t , may update from X_t but must be reported as a different information model.

Windows are non-overlapping. This prevents one raw observation from affecting multiple nominal decisions and makes payload accounting direct. A final short window receives its own integer quota. No model, reconstruction state, or window crosses an independent machine or simulation-run boundary.

3.2 PCA covariance-contribution score

After processing window t , incremental PCA gives eigenpairs (λ_i, v_i) for $i = 1, \dots, k$. We define

$$s_{t,j} = \sum_{i=1}^k \lambda_i v_{ij}^2. \quad (2)$$

This is the j th diagonal element of the rank- k approximation

$$C_k = \sum_{i=1}^k \lambda_i v_i v_i^\top. \quad (3)$$

It measures how much retained subspace variance is expressed through channel j . It is not a supervised measure of fault relevance and does not select one representative from a correlated pair. This limitation follows from the score definition.

The implemented method first normalizes the raw contributions:

$$u_{t,j} = s_{t,j} / \sum_l s_{t,l}. \quad (4)$$

If the total retained variance is zero or nonfinite, u_t is uniform. This fallback prevents a constant bootstrap from poisoning subsequent updates. For $0 \leq \rho < 1$, smoothing operates on normalized participation:

$$\bar{u}_t = \rho \bar{u}_{t-1} + (1 - \rho) u_t, \quad \bar{u}_1 = u_1. \quad (5)$$

Algorithm 1 Causal PCA-guided rate allocation

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1: Set  $r_{1,j} \leftarrow B$  for all  $j$ ; initialize PCA
2: for  $t = 1, 2, \dots$  do
3:   Announce  $r_t$  before observing window  $X_t$ 
4:   Convert  $r_t$  to integer quotas summing to  $\lfloor Bwd \rfloor$ 
5:   Acquire samples  $P_{\Omega_t}(X_t)$  under those quotas
6:   Causally reconstruct  $\hat{X}_t$  by zero-order hold
7:   Update incremental PCA using  $\hat{X}_t$ 
8:   Compute and smooth scores using Eq. (2)
9:   Allocate  $r_{t+1}$  using Eq. (6)
10: end for
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The special value $\rho = 1$ denotes the cumulative mean of the normalized score vectors. Normalization before smoothing prevents a high-energy window from receiving additional weight solely through its covariance trace; this differs from smoothing raw covariance contributions and then normalizing.

3.3 Rates, integer quotas, and acquisition

Set $p_{t,j} = \bar{u}_{t,j} / \sum_l \bar{u}_{t,l}$. The update after window t sets the next window’s target rates to

$$r_{t+1,j} = r_{\min} + p_{t,j}(Bd - r_{\min}d). \quad (6)$$

After clipping, excess is repeatedly redistributed among unsaturated channels until the feasible target budget is exhausted. We floor the target counts $wr_{t+1,j}$, then use largest remainders to reach exactly $\lfloor Bwd \rfloor$ samples. Positions are drawn without replacement. The minimum rate is a continuous target floor: integer rounding can reduce an individual realized rate below that floor, especially in a short final window. The aggregate cap is exact.

The receiver uses zero-order hold. Leading missing values use the last value from the preceding window; at cold start they use zero after train-fitted standardization. Linear interpolation is excluded from the primary online protocol because it requires future samples. The software retains it for explicitly offline use, but no such analysis is included in the primary or supplementary evidence here.

3.4 Complexity

For a window of length w , d channels, and k retained components, the incremental SVD update dominates computation. No controller-specific timing or microcontroller-readiness claim is made from operation counts alone. End-to-end claims require timing the acquisition, reconstruction, update, allocation, and communication-control path on named hardware.

3.5 AI assistance during revision

OpenAI Codex was used for code review and debugging, manuscript editing, and preparation of reproducibility and submission artifacts. Its assistance included changes

to evaluation code and generation of revised manuscript text. The authors retain responsibility for the analysis, claims, and final manuscript.

4 Properties and scope of the score

Proposition 1 (Covariance-diagonal identity) *For a positive semidefinite covariance matrix $C = \sum_{i=1}^d \lambda_i v_i v_i^\top$, the score in Eq. (2) satisfies*

$$s_j \geq 0, \quad \sum_{j=1}^d s_j = \sum_{i=1}^k \lambda_i, \quad (7)$$

$$C_{jj} - s_j = \sum_{i=k+1}^d \lambda_i v_{ij}^2 \geq 0. \quad (8)$$

For $k = d$, $s_j = C_{jj}$ and the raw PCA score equals the diagonal variance of the same covariance matrix.

Proof Non-negativity follows from $\lambda_i \geq 0$ and squared loadings. Summing over j and using $\sum_j v_{ij}^2 = 1$ gives the trace identity. Subtracting the truncated eigendecomposition from the full diagonal gives the residual. For $k = d$ the residual is zero. $\square$

This proposition supplies a rationale for the score but not an optimality claim. In particular, two exchangeable correlated channels receive equal scores. The previous claim that squared PCA loadings necessarily concentrate bandwidth on one member of a correlated pair is false and has been removed.

Proposition 2 *Assume $0 \leq r_{\min} \leq B$. Largest-remainder integerization of feasible target rates produces nonnegative quotas $q_{t,j} \leq w$ satisfying $\sum_j q_{t,j} = \lfloor Bwd \rfloor$ whenever that many samples are available.*

Proof Flooring produces an integer total no larger than the target. Each remaining unit is assigned only to a channel below its capacity. The procedure stops at the prescribed integer total, which is at most wd because $B \leq 1$. $\square$

If incremental PCA converges under a stationary distribution and its retained trace has a positive limit, the normalized score converges by continuity. For $\rho < 1$, its exponential average converges to the same limit; for $\rho = 1$, the cumulative mean converges by Cesàro convergence. No convergence claim is made for abrupt nonstationarity without additional assumptions.

5 Experimental protocol

Table 2 summarizes the dataset partitions and independent boundaries.

Table 2 Corrected dataset protocol. TEP counts are per fault type; SMD uses machine-1-1 and is not presented as the entire SMD collection.

Dataset	Channels	Train	Test	Boundary
TEP	52	5×500	5×500	simulation run
SMD-1-1	38	28,479	28,479	official stream
PSM	25	132,481	87,841	official stream

5.1 Datasets and splits

All 20 TEP fault types are evaluated. The 52 transmitted process channels include 41 measured and 11 manipulated variables; the budget counts values rather than assuming identical physical sensing costs. Simulations are split by simulation run; no window crosses a run, and online state is reset between runs [12, 13]. SMD and PSM use their official normal-training and labeled-test streams [14, 15]. Scaling is fit on training observations only. Random stratification of time points is not used. Hyperparameters are global unless selected exclusively from a validation subset on the training side.

TEP faulty trajectories contain a normal prefix followed by an introduced disturbance. Samples before onset are labeled normal instead of assigning the fault label to the entire trajectory. Both partitions use the released Training RData files: runs 1–5 train the classifier and runs 21–25 provide final evaluation for every fault type. The label convention is normal for sample indices below 20 and faulty from index 20 onward; it is recorded explicitly in the experiment manifest. SMD machine-1-1 and PSM retain their released order; normal training streams are not mixed with labeled test events. Using one SMD machine is explicit because concatenating machines would create artificial boundaries in a supposed single stream.

The aggregate MSL arrays used during an earlier revision are excluded from primary evidence. Their provenance does not establish the original entity boundaries; the source contains per-stream telemetry and encoded command inputs rather than 55 interchangeable physical sensors [16]. Treating the aggregate array as one continuous sensor stream would conflate independent histories and an inadequately specified communication model.

Allocation, PCA, and reconstruction state start fresh at every official train or test partition and every independent TEP run. The test stream therefore starts with a counted uniform bootstrap, rather than retaining part of the training controller’s state. Scaling and trained downstream detectors are retained from training. Full-rate training observations are available offline for the common scaler and fixed-detector reference. They are calibration inputs, not observations available to the online test allocator. Reported payload totals cover test acquisition; separate training ledgers are supplied. The budget therefore characterizes deployment replay, not total lifecycle acquisition cost. Partial final windows use the latest available rates; only complete windows update PCA.

Nonfinite numeric values are replaced before scaling. A StandardScaler is fit only to training observations and then applied without refitting to test data. The standardized value zero is consequently a train-derived mean and supplies the cold-start value

when a channel has not delivered an observation. Windows do not overlap, labels are unavailable to the allocator, and no parameter is selected from final test labels.

5.2 Comparators and downstream protocols

The primary comparators are Uniform allocation and two prior-window rules: Variance and change Threshold. All receive the same information, integer budget, bootstrap, and causal reconstruction. The primary study contains only these constrained rules and Full Data. Send-on-delta and supervised feature selection are discussed as related work, not represented as evaluated peers.

For labeled TEP runs, a random forest is evaluated in two ways: retrained for each reconstructed training set, and fixed after training on full-rate data [17]. Official normal-only anomaly-training datasets use an Isolation Forest with a threshold determined from training scores. The same retrained-versus-fixed distinction is reported. These protocols separate end-to-end adaptation from distortion of an existing detector.

Variance allocation computes per-channel variance from the preceding causally reconstructed window. Threshold allocation uses preceding-window mean absolute changes and prioritizes channels above the median change. Uniform allocation assigns the same target rate to every channel. All three use the same integer quota routine and zero-order hold as the PCA rule. Full Data is a reference, not a constrained competitor. Budget realization and reconstruction are held constant. However, PCA uses accumulated subspace information and smoothing, whereas Variance and Threshold use one preceding window. The comparison therefore evaluates complete allocation rules rather than isolating a PCA effect independently of temporal memory.

5.3 Scope of the comparisons with recent methods

The experimental question is whether the covariance-participation rule improves on elementary allocation rules when each method receives only previously reconstructed samples and must obey the same next-window integer sample cap. Uniform, Variance, and Threshold are transparent reference rules, not a claim to cover the state of the art. Full Data measures the cost of restricted acquisition and is not a budget-matched competitor.

We do not report numerical comparisons with CLARA, CoSS, or the DQN sampling method. The reasons, and the resulting limits on interpretation, are as follows.

1. CLARA’s clustering and fault-tolerance setting [8] would require an explicit mapping from network nodes and their relationships to benchmark channels, together with a network/failure model. Our benchmark protocol supplies neither. Removing these components would evaluate an adaptation, whose results could not be attributed to the complete method.
2. CoSS [9] learns sensor and rate rankings with a downstream classification network. A faithful TEP comparison would require a separate task-trained model, validation-only selection, and accounting for the data used during selection. The normal-only SMD and PSM training splits do not supply the labeled activity-classification task

used by CoSS. Such extensions are possible, but are not experiments performed in this study.

3. The DQN method [10] uses a state containing observations, remaining power, and sampling history, with a reward trading information against energy and redundancy. The present replay protocol has no calibrated power model and counts acquired scalar values rather than a reward penalty. A comparison would need to specify how states are formed from acquired values, charge all measurements used in learning and decisions, and constrain actions to the same per-window cap. We have not implemented or validated that adaptation.

These are scope-based reasons for the omitted experiments, not evidence that recent methods cannot be compared or would perform worse. They leave the empirical contribution limited to the implemented reference rules. In particular, TEP labels could support a separate supervised selection study, and clustered or learned policies could be assessed under an explicitly extended protocol. A broader comparative study remains necessary before claiming competitive performance against contemporary sensor-selection or learned-sampling systems.

5.4 Metrics and repetition

We report payload values and bytes, realized bandwidth, per-window budget excess, MAE, RMSE, normalized per-channel RMSE, precision, recall, F1, false positive rate, specificity, event recall, false-alarm events per 1000 samples, and detection delay. Point adjustment is not used. Experiments are repeated with five fixed joint seeds that control acquisition masks, detector randomness, and (for Isolation Forest) the fitting subset. Means and standard deviations therefore describe joint experimental variation, not acquisition alone or population uncertainty. Independent TEP simulation runs provide substantive replication; the official anomaly benchmarks are single-stream case studies, so no population-level inferential superiority is claimed for them.

For original and reconstructed streams X and $\hat{X}$, distortion is summarized by

$$\text{RMSE} = \sqrt{\frac{1}{nd} \sum_{t=1}^n \sum_{j=1}^d (X_{tj} - \hat{X}_{tj})^2}. \quad (9)$$

TEP RMSE pools squared errors over all test runs before taking the square root. Per-channel RMSE is normalized by the pooled test range and averaged; constant channels use a denominator of one. Test ranges are used only for retrospective error reporting, never for allocation, training, or threshold selection. An event is a maximal contiguous positive-label interval. Event recall counts an event when at least one unadjusted positive prediction falls inside it. Detection delay is measured from event start to the first such prediction. A predicted positive interval overlapping no true event is a false-alarm event. Each independent TEP run is scored separately for events before pooling counts and delay samples. Delay is conditional on detection and expressed in samples; missed-event counts are reported alongside it. If no event is detected, delay

Table 3 Corrected causal results at 50% bandwidth. Values are mean $\pm$ standard deviation across five joint experimental seeds. Full Data is shown as a detector reference. No point adjustment is used.

Dataset	Method	Retrained F1	Fixed F1	Event recall	RMSE
SMD-1-1	PCA-Triage	0.390 ± 0.017	0.401 ± 0.012	0.700 ± 0.168	9.024 ± 2.230
	Variance	0.407 ± 0.019	0.406 ± 0.011	0.825 ± 0.143	1.258 ± 0.246
	Threshold	0.393 ± 0.013	0.396 ± 0.013	0.725 ± 0.163	13.987 ± 17.107
	Uniform	0.395 ± 0.008	0.400 ± 0.010	0.850 ± 0.137	83.863 ± 19.805
	Full Data	0.409 ± 0.010	0.409 ± 0.010	0.875 ± 0.125	0.000 ± 0.000
PSM	PCA-Triage	0.031 ± 0.001	0.035 ± 0.007	0.139 ± 0.026	0.222 ± 0.011
	Variance	0.032 ± 0.001	0.035 ± 0.007	0.153 ± 0.022	0.193 ± 0.011
	Threshold	0.032 ± 0.002	0.034 ± 0.007	0.125 ± 0.028	0.228 ± 0.014
	Uniform	0.033 ± 0.004	0.035 ± 0.007	0.147 ± 0.032	0.237 ± 0.018
	Full Data	0.036 ± 0.007	0.036 ± 0.007	0.214 ± 0.016	0.000 ± 0.000

is undefined. Per-1000-sample alarm rates do not imply a common physical time scale across datasets. These definitions avoid retrospective point adjustment.

The global allocation configuration is $k = 10$, $w = 50$, $\rho = 0.95$, and $r_{\min} = 0.05$. Anomaly benchmarks use $B \in \{0.3, 0.5, 0.7\}$; the larger all-20-fault TEP experiment uses the primary comparison $B = 0.5$. Sampling seeds are 42, 123, 456, 789, and 1024. TEP uses a 50-tree class-balanced random forest. Normal-only benchmarks use a 100-tree Isolation Forest fitted to at most 20,000 normal training observations; its threshold is the 99th percentile of training anomaly scores. These values are fixed across datasets and are not selected from test outcomes.

Every acquisition call records the actual retained-mask count, available values, allowed integer quota, and excess. Results sum those ledgers, and a verifier independently reconciles every test total. Payload bytes assume four bytes per transmitted numeric value; they do not measure a network encoding or packet traffic. Full Data has a separately counted full observation mask.

Threshold sensitivity evaluates the predeclared training-score quantiles 0.95, 0.975, 0.99, 0.995, and 0.999 for every anomaly method and seed. The primary quantile remains 0.99; no test outcome chooses a preferred threshold. Complete point, event, distortion, and communication results for both detector protocols and all budgets appear in Online Resource 1 (ESM_1.pdf).

6 Results

Table 3 compares the anomaly benchmarks at 50% bandwidth; Table 4 varies the budget. Tables 5 and 6 report TEP aggregate and per-fault outcomes. Table 7 reports alarm and event outcomes.

The causal, mask-audited results depend on dataset, budget, and metric. All constrained methods meet their integer sample cap in every recorded window. Payload is an assumed four-byte representation, not measured network traffic.

At 50% bandwidth on SMD, PCA-Triage obtains point F1 0.390; the highest constrained mean is 0.407 (Variance). At 50% bandwidth on PSM, PCA-Triage obtains point F1 0.031; the highest constrained mean is 0.033 (Uniform). On all-20-fault TEP,

Table 4 Point-F1 sensitivity to the hard communication budget. The comparator column is the strongest non-PCA constrained method at that budget.

Dataset	Budget	PCA-Triage	Best comparator	F1
SMD	0.3	0.396	Variance	0.406
SMD	0.5	0.390	Variance	0.407
SMD	0.7	0.396	Threshold	0.409
PSM	0.3	0.030	Threshold	0.033
PSM	0.5	0.031	Uniform	0.033
PSM	0.7	0.032	Threshold	0.036

Table 5 Run-separated TEP results at 50% bandwidth, with the fault onset labeled within each run. Values are mean $\pm$ standard deviation across five joint experimental seeds.

Method	Retrained weighted F1	Fixed weighted F1	Macro F1	Binary F1	RMSE
PCA-Triage	0.675 $\pm$ 0.003	0.671 $\pm$ 0.004	0.684 $\pm$ 0.003	0.932 $\pm$ 0.002	0.270 $\pm$ 0.002
Variance	0.677 $\pm$ 0.002	0.673 $\pm$ 0.004	0.686 $\pm$ 0.003	0.931 $\pm$ 0.002	0.261 $\pm$ 0.002
Threshold	0.643 $\pm$ 0.005	0.644 $\pm$ 0.004	0.651 $\pm$ 0.005	0.924 $\pm$ 0.003	0.383 $\pm$ 0.002
Uniform	0.707 $\pm$ 0.003	0.686 $\pm$ 0.002	0.716 $\pm$ 0.003	0.937 $\pm$ 0.001	0.483 $\pm$ 0.001
Full Data	0.728 $\pm$ 0.001	0.728 $\pm$ 0.001	0.740 $\pm$ 0.002	0.958 $\pm$ 0.001	0.000 $\pm$ 0.000

PCA-Triage obtains weighted F1 0.675, versus 0.707 for Uniform and 0.728 for Full Data. No inferential statistical superiority is claimed. Complete distortion, payload, delay, false-alarm and threshold-sensitivity tables are supplied in the accompanying supplementary material.

7 Discussion

The PCA score should be interpreted as retained-subspace participation, not as a direct measure of fault relevance or conditional redundancy. Consequently, the method may be useful when dominant covariance structure aligns with measurement utility, but variance or uniform allocation may be preferable when low-variance channels contain rare diagnostic signals. Results are therefore reported by dataset, fault/event, and bandwidth rather than collapsed into a universal superiority statement.

The strict protocol updates PCA from reconstructed data and can accumulate feedback bias. Periodic exploration, missing-data subspace tracking, or a local full-rate gateway are possible extensions, but they change the information or communication model and require separate evaluation. Similarly, delayed linear interpolation can improve offline reconstruction but cannot support zero-delay monitoring claims.

The component count has a precise interpretation in the revised formulation. When $k = d$, the raw score is the marginal variance of the same accumulated covariance. This identity does not make the smoothed historical method identical to the preceding-window Variance baseline. For $k < d$, it discounts variance lying outside the retained subspace. This may remove nuisance variation, but it may also suppress an independent channel whose low-variance behaviour is safety critical. A minimum target rate reduces complete starvation in full windows but does not make the unsupervised

Table 6 Exact fault-type recall for all 20 TEP disturbances at 50% bandwidth, averaged across five joint experimental seeds. Pre-onset samples are labeled normal; confusing one fault for another is counted as incorrect here.

Fault	PCA-Triage	Variance	Threshold	Uniform	Full Data
1	0.981	0.980	0.983	0.980	0.986
2	0.969	0.970	0.970	0.970	0.973
3	0.158	0.155	0.113	0.167	0.145
4	0.958	0.945	0.952	0.975	0.977
5	0.740	0.719	0.624	0.765	0.918
6	0.997	0.997	0.996	0.997	0.998
7	0.995	0.995	0.995	0.995	0.998
8	0.779	0.785	0.737	0.814	0.814
9	0.136	0.131	0.103	0.154	0.234
10	0.495	0.532	0.400	0.663	0.721
11	0.707	0.704	0.639	0.690	0.734
12	0.768	0.757	0.746	0.774	0.790
13	0.686	0.699	0.638	0.708	0.726
14	0.947	0.953	0.963	0.955	0.962
15	0.134	0.125	0.109	0.157	0.303
16	0.353	0.392	0.238	0.552	0.632
17	0.806	0.818	0.715	0.832	0.863
18	0.811	0.815	0.776	0.831	0.838
19	0.507	0.507	0.476	0.628	0.809
20	0.653	0.659	0.743	0.654	0.698

score task-optimal. Conditional redundancy or supervised relevance would require a different score and a different claim.

Measurement and detection rankings need not agree. A strategy can reduce RMSE by preserving high-energy signals yet remove a small diagnostic transient, or increase RMSE while leaving a detector’s decision boundary intact. For this reason, the paper does not use F1 as a surrogate for measurement fidelity and does not use reconstruction error as proof of fault-detection utility. Both are reported alongside event recall and delay.

8 Limitations

The study uses replayed public data rather than a physical sensor network. The low full-rate anomaly F1 scores limit conclusions about practical diagnostic quality: the Isolation Forest is a fixed evaluation consumer, not an optimized anomaly detector. Threshold sensitivity exposes this dependence but does not establish deployment adequacy. Payload counts omit protocol headers unless explicitly added for a deployment. The public replay datasets provide no sensor-calibration uncertainty metadata, so the study reports reconstruction distortion but not a metrological uncertainty budget. PCA captures second-order linear structure only. The score is unsupervised and need

Table 7 Detection and alarm outcomes at 50% bandwidth with method-retrained detectors. TEP detection is binary (any fault), with event boundaries preserved within each run. Delay is conditional on detection; misses are reported separately. Entries are means and standard deviations across five joint experimental seeds.

Dataset	Method	Budget	Precision	Recall	FPR	Missed events	Delay (samples)	FA/1000
SMD-1-1	PCA-Triage	0.5	0.317 $\pm$ 0.013	0.507 $\pm$ 0.023	0.114 $\pm$ 0.003	2.4 $\pm$ 1.3	26.070 $\pm$ 17.873	6.159 $\pm$ 0.543
SMD-1-1	Variance	0.5	0.329 $\pm$ 0.016	0.535 $\pm$ 0.033	0.114 $\pm$ 0.008	1.4 $\pm$ 1.1	14.523 $\pm$ 15.970	6.988 $\pm$ 0.456
SMD-1-1	Threshold	0.5	0.320 $\pm$ 0.014	0.509 $\pm$ 0.017	0.113 $\pm$ 0.007	2.2 $\pm$ 1.3	28.300 $\pm$ 17.448	6.342 $\pm$ 1.036
SMD-1-1	Uniform	0.5	0.319 $\pm$ 0.008	0.518 $\pm$ 0.009	0.116 $\pm$ 0.003	1.2 $\pm$ 1.1	10.901 $\pm$ 7.975	5.337 $\pm$ 0.234
SMD-1-1	Full Data	1.0	0.330 $\pm$ 0.010	0.537 $\pm$ 0.011	0.114 $\pm$ 0.004	1.0 $\pm$ 1.0	4.967 $\pm$ 0.709	8.968 $\pm$ 0.552
PSM	PCA-Triage	0.5	0.976 $\pm$ 0.031	0.016 $\pm$ 0.001	0.000 $\pm$ 0.000	62.0 $\pm$ 1.9	906.807 $\pm$ 359.620	0.046 $\pm$ 0.046
PSM	Variance	0.5	0.987 $\pm$ 0.014	0.016 $\pm$ 0.001	0.000 $\pm$ 0.000	61.0 $\pm$ 1.6	952.049 $\pm$ 214.155	0.036 $\pm$ 0.025
PSM	Threshold	0.5	0.980 $\pm$ 0.012	0.016 $\pm$ 0.001	0.000 $\pm$ 0.000	63.0 $\pm$ 2.0	985.411 $\pm$ 323.244	0.064 $\pm$ 0.042
PSM	Uniform	0.5	0.974 $\pm$ 0.022	0.017 $\pm$ 0.002	0.000 $\pm$ 0.000	61.4 $\pm$ 2.3	678.599 $\pm$ 402.486	0.068 $\pm$ 0.055
PSM	Full Data	1.0	0.963 $\pm$ 0.026	0.018 $\pm$ 0.004	0.000 $\pm$ 0.000	56.6 $\pm$ 1.1	629.185 $\pm$ 95.826	0.155 $\pm$ 0.096
TEP	PCA-Triage	0.5	0.963 $\pm$ 0.001	0.904 $\pm$ 0.005	0.384 $\pm$ 0.012	0.0 $\pm$ 0.0	2.962 $\pm$ 0.879	7.223 $\pm$ 0.391
TEP	Variance	0.5	0.962 $\pm$ 0.001	0.903 $\pm$ 0.004	0.393 $\pm$ 0.009	0.0 $\pm$ 0.0	3.460 $\pm$ 0.554	7.170 $\pm$ 0.668
TEP	Threshold	0.5	0.965 $\pm$ 0.002	0.886 $\pm$ 0.007	0.354 $\pm$ 0.023	0.0 $\pm$ 0.0	4.588 $\pm$ 0.696	5.760 $\pm$ 0.398
TEP	Uniform	0.5	0.962 $\pm$ 0.001	0.914 $\pm$ 0.002	0.399 $\pm$ 0.011	0.0 $\pm$ 0.0	2.044 $\pm$ 0.438	9.672 $\pm$ 0.148
TEP	Full Data	1.0	0.943 $\pm$ 0.002	0.972 $\pm$ 0.002	0.638 $\pm$ 0.019	0.0 $\pm$ 0.0	0.350 $\pm$ 0.304	8.343 $\pm$ 1.114

not align with rare faults. Normal-only anomaly benchmarks require a detector different from supervised TEP classification. Hardware timing and energy measurements remain necessary before claiming embedded deployment readiness.

Internal validity is improved by causal timing, common reconstruction, common integer quotas, and train-only preprocessing, but results still depend on the chosen downstream detectors and global parameter setting. Construct validity is limited because payload values do not include packet headers, acknowledgments, clock synchronization, or control messages. External validity is limited by replayed benchmarks, one SMD machine, and the absence of a physical radio or industrial fieldbus. These limitations motivate hardware validation and multi-machine replication rather than stronger deployment language.

9 Conclusion

We formulated PCA-guided sensor-rate allocation as a causal measurement problem with an explicit information pattern and hard per-window budget. The revised method uses the diagonal of a rank- k covariance approximation, makes next-window decisions from past reconstructed observations, and avoids future-value interpolation. Its value must be judged jointly by measurement distortion, realized communication, event detection, delay, and false alarms. The resulting formulation supports a narrower but testable contribution and clearly identifies settings in which simpler allocation rules may be better.

Data and code availability

The accompanying source archive contains the causal pipeline, official-partition loaders, fixed seeds, regression tests, source/data SHA-256 manifests, measured per-window acquisition ledgers, and scripts that regenerate all tables. The accompanying archive defines the evaluated revision; the development repository is <https://github.com/ankitlade12/pca-sensor-triage>. The archive (Online Resource 2, ESM_2.tar.gz) identifies the evaluated revision; the development repository may contain a different version. Public data are obtained from their original repositories under their respective terms: TEP at <https://doi.org/10.7910/DVN/6C3JR1>, SMD at <https://github.com/NetManAIops/OmniAnomaly>, and PSM at <https://github.com/eBay/RANSynCoders>.

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Competing interests

On behalf of all authors, the corresponding author states that there is no conflict of interest.

Author contributions

All authors contributed equally.

Ethics approval

Not applicable. This study uses public process simulation and server benchmark data and introduces no human-participant or animal experiments.

Consent to participate

Not applicable.

Consent for publication

Not applicable; no identifiable human participant material is included.

Prior dissemination

An earlier version is available as <https://arxiv.org/abs/2604.05045>. The present manuscript corrects the evaluation protocol and supersedes that version’s numerical results and claims.

Supplementary information

Online Resource 1 (ESM_1.pdf): Complete detection, reconstruction, payload, event and threshold-sensitivity results for both detector protocols.

Online Resource 2 (ESM_2.tar.gz): Evaluated source, tests, experiment results, acquisition ledgers, provenance manifests and reproduction instructions. Raw third-party datasets are obtained separately from their original sources.

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